

DEGREE: MSc Artificial Intelligence

Module: Ethics and Legal Framework in AI

Assignment Title: Ethical and Legal Analysis of AI in Biotech: A Comprehensive Business Assessment for PARACELUSUS

Assignment Type: Essay

Word Limit: 3000 words (+/- 300)

Weighting: 100%

Issue Date: 24/04/2026

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Feedback Date: 29/06/2026

Plagiarism:

When submitting work for assessment, students should be aware of the InterActive/Canvas guidance and regulations in concerning plagiarism. All submissions should be your own, original work.

You must submit an electronic copy of your work. Your submission will be electronically checked.

Learner declaration		
Word count		
Use of proof-reader/proof-reading service (see guidance contained in Academic Misconduct Regulations on appropriate use)	YES	NO
If 'yes', please specify which service (e.g. Grammarly, Studiosity etc.)		

I confirm that I submit this work as my own work and that I have cited all sources I have used, and I understand that using sources without citing them correctly may be considered as Academic Misconduct.	YES	NO
I confirm that I have followed guidance on the acceptable use of AI tools for this assignment where such guidance has been issued by my tutors.	YES	NO
Where use of appropriately cited use of AI tools is permitted, I confirm that I have cited in accordance with the UCA Harvard Referencing Standard.	YES	NO

Harvard Referencing:

The Harvard Referencing System must be used. The Wikipedia, UKEssays.com or similar websites must **not** be used or referenced in your work.

By submitting this assessment, you are confirming that you have understood and are complying with the University's Academic Misconduct Regulations.

Plagiarism Detection (Turnitin)

BSBI uses Turnitin (an on-line text matching tool) to help identify cases of plagiarism. Turnitin will compare a submitted assignment with a database of journal articles, books, web pages and previously submitted assignments.

Signature (Student): _____ Date: _____

Introduction

AI-powered tools are reshaping the biomedical landscape by enabling better patient

outcomes. However, this disruption also brings significant ethical and legal concerns around privacy, bias, transparency, and consent.

This assignment tasks you with conducting a comprehensive ethical and legal assessment of a fictitious Biotech company, PARACELUSUS, to ensure its AI products meet ethical standards and comply with jurisdiction-specific legal frameworks.

Learning Outcomes:

LO1. Learn to critically evaluate the ethical challenges and legal requirements associated with AI applications in the biotechnology and healthcare sectors, particularly in relation to privacy, fairness, autonomy, and accountability.

LO2. Learn to synthesize ethical principles and legal standards to design actionable mitigation strategies and provide practical, evidence-based recommendations for ensuring compliance and ethical responsibility in AI-driven products.

LO3. Learn to apply relevant ethical theories and legal frameworks to hypothetical scenarios, justifying decisions and proposals for responsible AI deployment.

Assessment Criteria: Weighting 100%

3000 words

Scenario:

The (fictitious) biotech sequencing company, *PARACELUSUS*, is expanding its services and plans to offer a range of 5 new genetic services. These services will use AI to predict genetic diseases, complex morbidity, personal attitudes, skills, and psychological profiles, each constituting a distinct product:

1. *PARACELUSUS GeneGuard*, which signals protection and early warning against inherited diseases.
2. *PARACELUSUS PolyRisk*, which emphasizes layered, interconnected risk modeling using AI.
3. *PARACELUSUS PersonaMap*, which suggests a mapping of personality tendencies in a neutral, exploratory way.
4. *PARACELUSUS AptiCore*, which is performance-oriented and practical, useful for education or career guidance.
5. *PARACELUSUS PsycheDecode*, blending scientific credibility with a sense of deep insight into cognition and psychology.

These services will be marketed to both individuals and businesses under the following

arrangements:

- **For Individuals:** Private customers can submit their samples and data to receive their personal results at home, depending on the specific product(s) they have purchased.
- **For Businesses:** Employees can voluntarily submit their samples and data. Employees will receive their full individual results based on the product(s) purchased by their employer, while employers will have access to aggregated results only.

Your Role:

You are an **AI Ethics and Regulatory Consultant** hired by *PARACELSUS* to conduct a comprehensive assessment of the ethical and legal implications of their new products, *in a single jurisdiction of your choice*. Your task is to write a **professional Business Assessment and Recommendations Report** to help the company ensure ethically responsible innovation and legal compliance.

Your Deliverable:

You are required to produce an **Essay** in the form of a **Business Assessment and Recommendations Report**. This document should be presented as if it were an official deliverable to the client, *PARACELSUS*.

Tasks:

1. Preparation:

- **Synthesize** the various aspects of the problem. Take time to analyse the scenario critically, researching similar companies and their approach to addressing ethical and legal concerns in the AI-driven biotech space.
- Select **one jurisdiction** for your analysis (e.g., EU, US, UK, Japan, India), and study the respective legal and regulatory frameworks concerning AI, healthcare, and data privacy.
- **Identify** emerging ethical and regulatory concerns.
- Ensure your report draws upon **empirical research** and established **ethical frameworks** in AI and biotechnology.

2. Report Content:

A. Overview of Ethical and Legal Issues

- **Evaluate** the broad ethical and legal implications associated with the proposed AI products. **Identify and appraise** potential ethical challenges such as:
 - **Privacy and data protection** concerns, particularly regarding genetic and health data.
 - **Bias and fairness** in AI algorithms, especially regarding potential discriminatory outcomes based on genetic information.
 - **Autonomy and consent**, particularly in scenarios where employees are asked to voluntarily submit their data.

- **Accountability and transparency** regarding AI decision-making processes.
- **Assess** the potential impact on different **stakeholders**, which you should clearly identify. Discuss how each group may be affected by these products, ethically and legally.

B. In-Depth Ethical Analysis

- **Examine and critically evaluate** the ethical implications, categorizing them under key themes (e.g., privacy, fairness, autonomy, transparency).
- Provide **examples of potential ethical dilemmas** or adverse incidents that may arise, such as breaches of data privacy or biased outcomes from the AI models.
- **Integrate** existing ethical theories and frameworks into your analysis, demonstrating how these frameworks apply to the scenario.

C. In-Depth Legal Analysis

- Based on your chosen jurisdiction, **analyse** the relevant legal frameworks, such as data protection laws (e.g., GDPR in the EU), healthcare regulations, and AI-specific policies.
- **Identify and assess** specific legal challenges that *PARACELSUS* products may face. Consider issues related to data processing, informed consent, liability for AI decisions, and regulatory compliance.
- **Evaluate** the adequacy of current legal frameworks in addressing the ethical issues identified in your analysis and discuss where legal gaps may exist.

D. Mitigation Strategies

- **Propose** practical, ethical, and legally compliant **design strategies**. These strategies may include:
 - **Developing transparent AI models** that can be explained to both customers and regulatory bodies.
 - **Ensuring fair and unbiased outcomes** by improving the diversity and representativeness of training data.
 - **Enhancing data protection measures** to safeguard sensitive genetic and health data.
 - **Establishing clear accountability mechanisms** for the use and outcomes of AI systems.
- Discuss the possibility of **removing or modifying specific products** from *PARACELSUS*'s portfolio if the ethical or legal risks cannot be sufficiently mitigated.

E. Recommendations

- **Synthesize** your findings into 5-7 of actionable recommendations for *PARACELSUS*. These recommendations should be practical, evidence-based, and aligned with both ethical principles and legal requirements.
- Align advice with both ethical norms and legal compliance expectations.
- Advise on internal governance (e.g., periodic review, cross-functional teams, third-party oversight).
- Suggest strategies to build trust (e.g., voluntary transparency reports, user education).

3. Report Format and Structure

Your business report **must include:**

- Cover page with title and professional look (right after the BSBI template front matter)
- Executive summary
- Table of contents
- Main Body (A–E)
- Conclusion
- References (Harvard style)

Support your analysis and recommendations with citations from the relevant literature, ethical guidelines, and legal regulations.

Ensure that your arguments are **coherent, well-structured, and logically presented**, using **formal language** and a **professional outline**, appropriate for a consultancy report.

Checklist for Submission:

- **Is your report structured** as a professional business document?
- **Have you critically analysed** the ethical and legal implications of the products?
- **Have you referenced** relevant legal frameworks, academic literature, and ethical theories?
- **Are your recommendations practical, evidence-based, and feasible** for the company to implement?

Grading Criteria:

- Overview (20%)
- In-depth ethical analysis (10%)
- In-depth legal analysis (20%)
- Mitigation strategies (20%)
- Recommendations (20%)
- Report format and structure (10%)

Students are encouraged to explore additional reputable sources in the digital library beyond the list provided to strengthen their individual research and critical analysis skills. The unit lecturer can help with recommending additional literature.

The criteria for successfully passing the units with two assessment components require passing both components successfully. Passing marks for BSc and MA/MSc levels are different and are stated at the end of the assignment brief. Please find the examples below:

- **Example 1** (e.g., pass mark: 50)
- Component 1 (50% weighting) - 55
- Component 2 (50% weighting) - 55

- Final average: 55
- Result: **Pass (both components passed and therefore the unit is passed)**

- **Example 2** (e.g., pass mark: 50)
- Component 1 (50% weighting) – 80
- Component 2 (50% weighting) – 30
- Final average: 55
- Result: **Fail (because although the average is above 50, one of the components is a failure)**

GRADING DESCRIPTORS: LEVEL 7

EXPERIMENTATION & INNOVATION								
	FAIL			PASS				
Threshold Criteria	0-29%	30-39%	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Deals with complex issues both systematically and creatively demonstrating self-direction and originality in tackling and solving problems	Little to no ability to use techniques to deal with complex issues systematically (including those of ethics and sustainability) and creatively to solve problems and/or make decisions.	Low utilisation of established techniques to deal with complex issues systematically (including those of ethics and sustainability) and creatively to solve problems and/or make decisions, but with limitations in techniques or approach.	Limited research or advanced scholarship to their area of study by using a range of information and established and advanced techniques	Competent understanding of solving problems, through own research or advanced scholarship displaying a comprehensive understanding of established and advanced techniques	Good understanding of solving problems through own research and advanced scholarship critically selecting and displaying a comprehensive understanding of established and advanced techniques.	Very Good problem-solving skills displaying a comprehensive understanding of techniques applicable to their own research or advanced scholarship	Excellent range of extremely well-developed problem-solving displaying an understanding of techniques applicable to their own research or advanced scholarship beyond which is taught.	Exceptional problem-solving skills with sophisticated evaluation and application of a wide range of advanced information and techniques to undertake projects.
Comprehensive understanding of techniques applicable to their own research or advanced scholarship	Little to no understanding of techniques applicable to their own research or advanced scholarship or their limitations and ambiguities.	Low understanding of techniques applicable to their own research or advanced scholarship including their limitations and ambiguities.	Limited understanding of key techniques applicable to their own research or advanced scholarship including their limitations and ambiguities.	Competent understanding of techniques applicable to their own research or advanced scholarship including their limitations and ambiguities	Good understanding of techniques applicable to their own research or advanced scholarship and a some understanding of more specialised techniques.	Very good understanding of techniques applicable to their own research or advanced scholarship and a some understanding of more specialised techniques.	Excellent understanding of techniques applicable to their own research or advanced scholarship and mastery of some more specialised areas.	Exceptional understanding of techniques applicable to their own research or advanced scholarship and mastery of some more specialised areas.

GRADING DESCRIPTORS: LEVEL 7

RESEARCH & ANALYSIS								
	FAIL			PASS				
Threshold Criteria	0-29%	30-39%	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Systematic understanding of knowledge, and a critical awareness of current problems and/or new insights, much of which is at, or informed by, the forefront of their academic discipline, field of study or area of professional practice	Little to no knowledge of the subject with limited breadth or depth or deficiencies in major areas or currency.	Low knowledge of the subject lacking coherence, breadth, or detail with only some reference to ideas or arguments at the forefront of any part of the subject.	Limited knowledge to deal with terminology, facts and concepts some of which is informed by the forefront of defined areas of the subject.	Competent knowledge of ideas or arguments at the forefront of any part of the subject sufficient to deal with current issues in the discipline, generally more descriptive than critical or analytical.	Good knowledge of ideas or arguments at the forefront of any part of the subject showing a clear, critical insight into the discipline as whole and current issues/problems.	Very good knowledge of ideas or arguments at the forefront of the subject some of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.	Excellent knowledge of ideas or arguments at the forefront of the subject many of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.	Exceptional knowledge of ideas or arguments at the forefront of the subject most of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.
Conceptual understanding that enables the student to display originality in the application of knowledge	Little to no conceptual understanding or argument and a focus on descriptive explanations which do not comment on arguments of others or alternative views.	Low conceptual understanding and arguments are weak or poorly constructed, and the work does not critically evaluate the arguments of others or consider alternative views.	Limited conceptual understanding and argument construction with critical evaluation of alternative views or comment on advanced scholarship.	Competent conceptual understanding and argument construction with critical evaluation of a range of views and consistent engagement with advanced scholarship.	Good conceptual understanding which critically evaluate and synthesise other views and information with a thoughtful interpretation of advanced scholarship.	Very good conceptual understanding which systematically synthesises a wide range of views with a critical insight into advanced scholarship.	Excellent conceptual understanding which critically apply a wide range of views through a perceptive use of advanced scholarship.	Exceptional conceptual understanding of publishable quality with systematic engagement and usage of advanced scholarship.

GRADING DESCRIPTORS: LEVEL 7

ENGAGING WITH PRACTICE								
	FAIL			PASS				
Threshold Criteria	0-29%	30-39%	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Practical understanding of how established techniques of research and enquiry are used to create and interpret knowledge in the discipline	Little to no evidence of background investigation, analysis, research, enquiry, ethical awareness, and/or study.	Low evidence of background investigation, analysis, research, enquiry, ethical awareness, and/or study.	Limited background investigation, analysis, research, enquiry, ethical awareness, and/or study using established techniques, with the ability to extract relevant points.	Competent investigation, analysis, research, enquiry, ethical awareness, and/or study using established techniques accurately, and can critically appraise and use academic sources.	Good background investigation, analysis, research, enquiry, ethical awareness, and/or study using established techniques accurately, and possesses a well-developed ability to critically appraise a wide range of sources.	Very good, independent, extensive and appropriate investigation, analysis, research, enquiry, ethical awareness, and/or study beyond the usual range, and critically evaluates this to advance the work and/or direct arguments.	Excellent independent, extensive and appropriate investigation, analysis, research, enquiry, ethical awareness, and/or study well beyond the usual range, and critically evaluates this to advance the work and/or direct arguments.	Exceptional investigation, analysis, research, enquiry, ethical awareness, and/or study which demonstrates carefully considered depth and breadth and critically synthesises this to advance the work and/or direct arguments.
Originality in the application of knowledge	Little to no technical, creative or artistic skills related to their area of study.	Low technical, creative or artistic skills related to their area of study.	Limited technical, creative or artistic skills required for area of study.	Competent technical, creative or artistic skills required for area of study.	Good technical, creative or artistic skills required for area of study.	Very good range of technical, creative or artistic skills.	Excellent range of technical, creative or artistic skills	Exceptional range of technical, creative or artistic skills
Independently advance your own knowledge and understanding, and to develop new skills to a high level.	Little to no contribution to group activity and/or undertaking further training at a high/advanced level.	Low contribution to group activity and/or undertaking further training at a high/advanced level.	Limited contribution to group activity and/or undertaking further training at a high/advanced level.	Competent contribution to group activity and/or independently undertakes further training at a high/advanced level.	Good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Very good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Excellent contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and leadership	Exceptional contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and strong leadership.

GRADING DESCRIPTORS: LEVEL 7

REALISATION & COMMUNICATION								
	FAIL			PASS				
Threshold Criteria	0-29%	30-39%	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Communicate information, ideas, problems and solutions to both specialist and non-specialist audiences.	Little to no clarity in the communication of ideas, problems and solutions to audiences.	Low clarity in the communication of ideas, problems and solutions to audiences.	Limited clarity in the communication of ideas, problems and solutions to audiences.	Competent communication of ideas, problems and solutions to audiences.	Good, confident and clear communication of ideas, problems and solutions to audiences in a range of means / media.	Very good, confident and clear communication of ideas, problems and solutions to audiences in a range of means / media.	Excellent communication of ideas, problems and solutions to audiences in a range of means / media.	Exceptional communication of ideas, problems and solutions to audiences in a range of means / media.

GRADING DESCRIPTORS: LEVEL 7

PERSONAL & PROFESSIONAL CONNECTIVITY								
	FAIL			PASS				
Threshold Criteria	0-29%	30-39%	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Independently advance your own knowledge and understanding, and develop new skills to a high level.	Little to no contribution to group activity and/or undertaking further training at a high/advanced level.	Low contribution to group activity and/or undertaking further training at a high/advanced level.	Limited contribution to group activity and/or undertaking further training at a high/advanced level.	Competent contribution to group activity and/or independently undertakes further training at a high/advanced level.	Good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Very good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Excellent contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and leadership	Exceptional contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and strong leadership.
Qualities and transferable skills necessary for employment requiring: (a) the exercise of initiative, ethical and personal responsibility (b) decision-making in complex and unpredictable contexts	Little to no ability to manage learning and/or exercise initiative, ethical and personal responsibility and/or decision-making in complex and unpredictable situations	Low ability to manage learning and/or exercise initiative, ethical and personal responsibility and/or decision-making in complex and unpredictable situations	Limited ability to manage learning and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Competent ability to manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Good ability to systematically manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Very good ability to systematically manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations.	Excellent ability to manage learning on own initiative, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Exceptional ability to manage learning on own initiative, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations
	Little to no use of appropriate terminology, limited vocabulary and many errors in spelling, grammar and syntax.	Low use of appropriate terminology, with many errors in spelling, vocabulary and syntax.	Limited expression, style and appropriate vocabulary with errors in spelling, grammar and syntax which affect understanding.	Competent expression, style, and appropriate vocabulary with some errors in spelling, grammar and syntax which do not affect understanding.	Good expression, style and appropriate vocabulary with some errors in spelling, grammar and syntax.	Very good expression, style and appropriate vocabulary with minimal errors in spelling, grammar and syntax.	Excellent expression, style and appropriate vocabulary with minimal errors in spelling, grammar and syntax.	Exceptional expression, style and appropriate vocabulary with no errors in spelling, grammar and syntax.

GRADING DESCRIPTORS: LEVEL 7

	Little to no evidence of basic numeracy or digital literacy, hardware and software skills	Low evidence of basic numeracy or digital literacy, hardware and software skills competency.	Limited evidence of numeracy or digital literacy, hardware and software skills competency.	Adequate evidence of numeracy or digital literacy, hardware and software skills competency.	Good evidence of numeracy or digital literacy, hardware and software skills competency.	Very good evidence of numeracy or digital literacy, hardware and software skills	Excellent evidence of numeracy or digital literacy, hardware and software skills competency.	Exceptional evidence of numeracy or digital literacy, hardware and software skills competency.
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	competency.					competency.		
	Does not demonstrate achievement of professional competence when assessed against the requirements of a professional, statutory or regulatory body (PSRB).			The student has demonstrated achievement of professional competence when assessed against the requirements of a PSRB.				
	Inaccurate use of terminology with limited vocabulary and many errors in spelling, grammar and syntax. Inaccurate terminology, with many errors in spelling, vocabulary and syntax.			The student has adhered to the appropriate rules and/or conventions set by regulators or the industry.				